Alex Uamai

Mechatronics engineer building robotic machines for food production. Incoming M.S. Robotics, Johns Hopkins.

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PROFILE

I take electromechanical hardware from a blank SolidWorks file to the factory floor, validated with FEA and real testing. I've founded and scaled a startup to a 12-person team, and I'm now building robot-learning skills in simulation with an SO-101 arm in MuJoCo.

EXPERIENCE

Unifiller Systems · Mechatronics Engineering Project Leader, R&D — Dec 2025 – Present

- Own the mechanical design of an industrial robotic food-automation platform, from concept through prototype validation and production support
- Design end-effectors, precision assemblies, tooling and fixtures in SolidWorks, using GD&T and tolerance analysis to keep them manufacturable
- Validate designs with FEA and first-principles analysis before build, then prototype, test and iterate on hardware results
- Integrate motion, sensing, pneumatics and embedded electronics into one platform; drive DFM/DFA, sourcing, drawings, BOMs and change control

Grubchain · Founder & CEO — Sep – Dec 2025

- Founded a stablecoin payments company and grew the team from just me to 12
- Led product and engineering on a blockchain payment rail connecting merchants and consumers, with a focus on the Nigerian market
- Represented the company on Channels Television (Sep 2025) and in a Vantech Journal op-ed (Jun 2025)

Bethel Systems · Technical & Business Analyst — Dec 2024 – Dec 2025

- Turned early-stage client product ideas into engineering development plans, including support for patent applications

VMAC Air · Mechanical Engineering Intern — Sep – Dec 2021

- Designed manufacturing fixtures and production test equipment in SolidWorks for manufacturing improvement projects

Ziva Dynamics · Software Developer Intern — May – Aug 2021

- Shipped production C++ and Python with a Qt/PySide2 interface, and contributed to Git-based automated testing

Mulgrave School · IT Intern — Jan – May 2021

- Built a Raspberry Pi remote 3D-printing system and administered Windows Active Directory

PROJECTS

Poundo Robot · Independent build — Oct 2026 – Present

- Building a robot arm that learns to turn poundo yam from video; simulating an SO-101 arm in MuJoCo with Jacobian IK, PD control and contact-force sensing

Two-wheel self-balancing robot

- Modeled dynamics and designed a state-space feedback controller in MATLAB/Simulink, verified in a 3D Simscape simulation

Lief, foot wearable for dystonia pain — Capstone

- Led controls: PWM vibration-motor firmware, the protoboard circuit and a desktop GUI for session tracking

Embedded controllers

- STM32 tilt game controller (SPI accelerometer driver, interrupts, USB streaming); BLE control board brought up with J-Link; 8051 PWM motor controller with encoder feedback

EDUCATION

Johns Hopkins University · M.S., Robotics & Autonomous Systems Engineering — Starts Spring 2027

Simon Fraser University · B.A.Sc., Mechatronic Systems Engineering — 2018 – 2023

SKILLS

Robotics & sim	MuJoCo, LeRobot, SO-101, inverse kinematics, PD control, PyTorch, industrial robotics, motion control
Mechanical	SolidWorks, GD&T, tolerance analysis, FEA, DFM/DFA, drawings & BOMs, rapid prototyping, end-effector design
Electromech.	Actuators, sensors, pneumatics, embedded systems, PCB integration, BLE, STM32
Software	Python, C++, C, MATLAB/Simulink, Qt, Git, Linux, Jenkins, Jira, Linear, change management